Maanya Chugh

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EDUCATION

University of Texas at Austin **Expected Graduation: Winter 2027**

Bachelor of Science in Information Technology (Honors) ***GPA: 3.8/4.0***

- **Specialization:** Human-Centered Data Science
  Relevant Coursework: Data Structures, Applied Machine Learning with Python, Text Mining & NLP Essentials, Mobile Application Development, Virtual Reality & Game Design, Ethical AI, UI/UX Design

Danish Institute for Study Abroad - Copenhagen, Denmark **August - December 2024**

UT Global Ambassador

- Selected as 1/30 to conduct hands-on research in Scandinavian health systems; proposed tech-driven care solutions.

TECHNICAL SKILLS:

Languages: Python, TypeScript, JavaScript, Java, C++, SQL, HTML/CSS

Frameworks & Tools: PyTorch, TensorFlow, Hugging Face, Scikit-learn, RAG, Agno, MCP, FastAPI, Flask, Streamlit, React, PostgreSQL/pgvector, MongoDB, Docker, GitHub Actions, Langfuse, OpenTelemetry, Prometheus, Nginx, Git

EXPERIENCE

Tesla - Fremont, CA **August 2026 - Present**

AI Engineer Product Manager Intern – Supply Chain and Process Systems

- Scaled an internal supply-chain AI platform from **50 to 400 users**, redesigning production RAG ingestion with structure-aware chunking and hybrid retrieval.
- Built **LLM-as-a-judge evaluation pipelines** for grounding, retrieval, and tool use, improving agent performance from **36% to 64%**, with subsequent fixes reaching **79%**.
- Implemented a **Dream-RSI-inspired retrieval optimization loop**, raising held-out retrieval scores from 0.761 to 0.838 **(+10.1%)**, and targeted embedding updates that cut knowledge refreshes from **days to minutes**.

Bank of America – Pennington, NJ **May 2026 - August 2026**

SWE Intern – Global Technology

- **Architected & deployed** an enterprise Retrieval-Augmented Generation (RAG) platform, ingesting **1,000+ multi-format internal assets** (PDFs, spreadsheets) to generate vector embeddings & leveraging MongoDB vector search to enable AI-powered retrieval & **automate 40% of manual tasks**.
- **Automated MongoDB operational workflows & data-processing pipelines**, improving database platform scalability & **reducing operational latency by 25%**.

Soundscape-to-Image Lab – UT Austin **August 2025 - Present**

Undergraduate Researcher – Multimodal AI

- **Designed & ran multimodal experiments across nearly 12,000 samples**, evaluating audio, visual, text, & spatial features through ablation studies & **achieving an R^2 of 0.63**.
- **Fine-tuned models & optimized feature pipelines, hyperparameters, & training configurations**, improving predictive performance by **10%** & built automated preprocessing & evaluation workflows for data ingestion, feature extraction, experimentation, & reproducible result analysis.

Depository Trust Clearing Corporation (DTCC) – Jersey City, NJ **May 2025 - August 2025**

IT Intern – Internal Audit & Data Analytics

- **Engineered Python & KNIME automation workflows**, cutting audit cycle times by **25%** and standardizing processes across **3 internal teams**.
- **Designed & deployed ML and analytics solutions**, achieving **86% model accuracy** and building a centralized KNIME Business Hub application supporting **50+ users**.

TECHNICAL PROJECTS:

AI-Driven Cardiac Risk Classifier | Python, Scikit-learn, ML Pipelines

- **Engineered ML pipelines on 10K+ Electronic Health Records (EHRs)**, applying feature engineering to achieve **85% classification accuracy** using Logistic Regression & Random Forest models.
- **Optimized reproducible data preprocessing pipelines**, reducing execution time by **30%**.

SymptomSync – Clinical NLP Triage Assistant | Python, NLP, Data Structures

- **Developed NLP algorithms for clinical symptom extraction & automated triage**, supporting clinic decision-making workflows.
- **Optimized query matching using efficient data structures**, reducing response latency by **36%**.

LEADERSHIP & ACTIVITIES

Hook 'Em Hacks — *Founder & President* **September 2025 - Present**

- **Founded & led UT Austin's inaugural AI-focused hackathon** with **250+ participants**, managing a **$10K budget**, securing corporate sponsorships, & directing event operations.